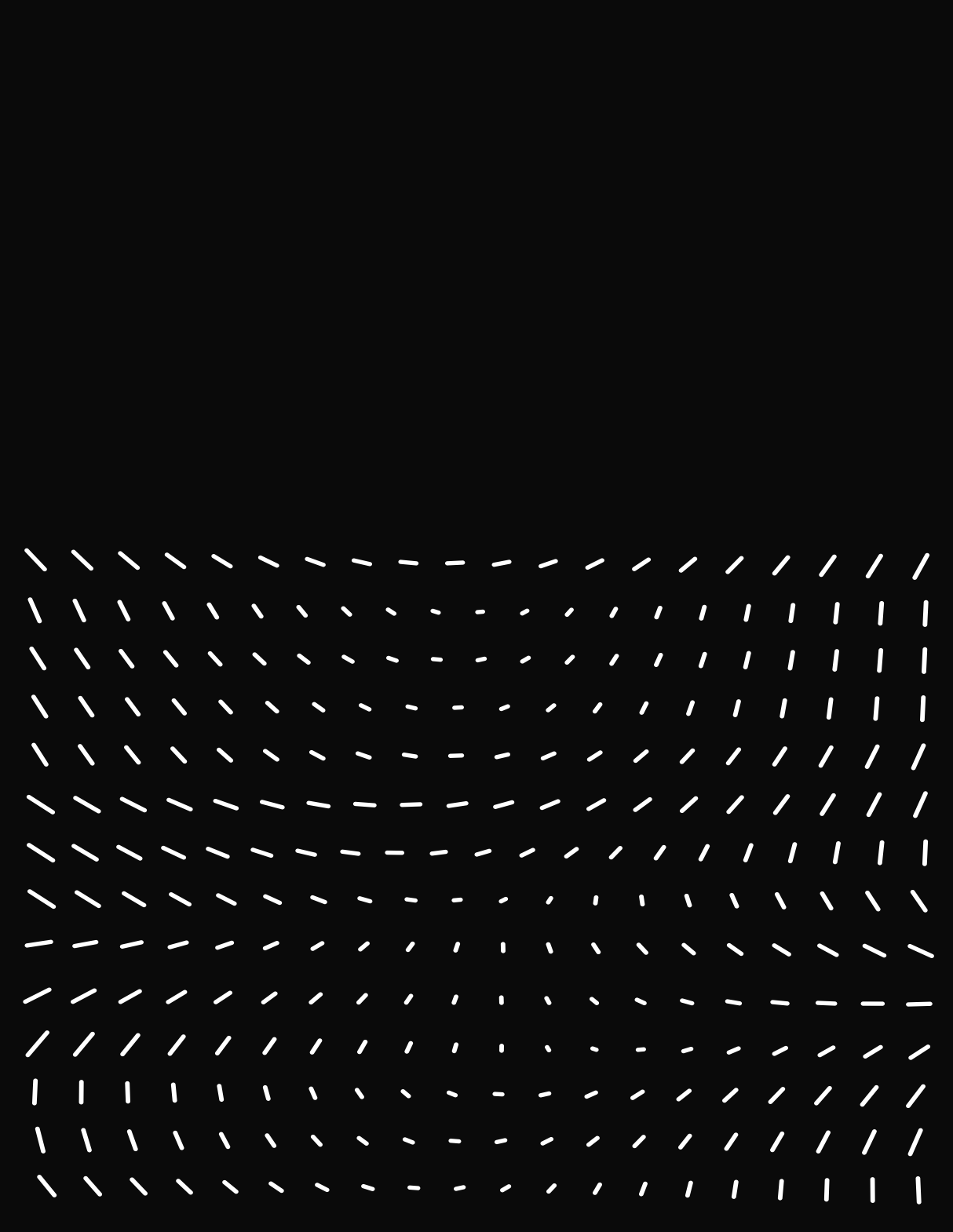


CT-1 (2021)

Q1. What are the main reasons to use microcontroller? What is the maximum clock speed for ATMega16 and how it can be implemented at ATMega16 microcontroller?

Main reasons to use MC

i) Number of pins less than MP

ii) Design time decreased

iii) Board layout size decreased

iv) Cost effective

v) CPU, RAM, I/O and timer on the same chip

vi) Can be used for specific purpose

vii) Reliability greater than MP

- The maximum clock speed for ATmega 16 controller is 16 MHz
- It can be implemented using an external crystal oscillator
- XTAL1 and XTAL2 are the external oscillator pin in ATmega16
- There are 32-8 bit general purpose registers in Atmega 16 mc

Suppose, you are given an analog quantization size of 0.75 Volt, where V_{min}=0 Volt and V_{max}=12 Volt. Calculate the desired number of bits for an A/D converter. **05**

$$Q = (V_{max} - V_{min})/N$$

$$\Rightarrow 0.75 = (12-0)/N$$

$$\Rightarrow N = 16$$

here N is the number of possible states

$$N = 2^n$$

$$\Rightarrow n = \log_2(N) = \log_2(16)$$

$$= 4$$

So, desired number of bits are 4

Suppose, it is given V_{in} = 0.75 Volt, V_{ref} = 1 Volt and 8-bit of resolution for a Successive Approximation A/D converter. Find a 8-bit digital output for the given V_{in}.

1	0	1	1	1	1	1	1

Vin = 0.75 volt
Vref = 1 volt

I) Vref x 1/2 = 0.5, Vin > V, MSB = on

II) 0.5 + 1/4 = 0.75

V = Vin, MSB1 = off

III) 0.5 + 1/8 = 0.625

Vin > V, MSB2 = on

IV) 0.5 + 1/8 + 1/16 = 0.6875

Vin > V, MSB3 = on

V) 0.5 + 1/8 + 1/16 + 1/32 = 0.71875

Vin > V, MSB4 = on

Similarly all others will be 1

Mid-20

b) Differentiate between Tiny, Mega and Xmega AVR microcontrollers. Based on the size of the flash memory, ATMega16 and ATMega328 microcontrollers belong to which category?

	# of pins	Flash memory	Special features
Tiny	6-32	0.5-8 KB	Small in size
Mega	28-100	4-256 KB	Extended peripherals
Xmega	44-100	16-384 KB	DMA, event system included

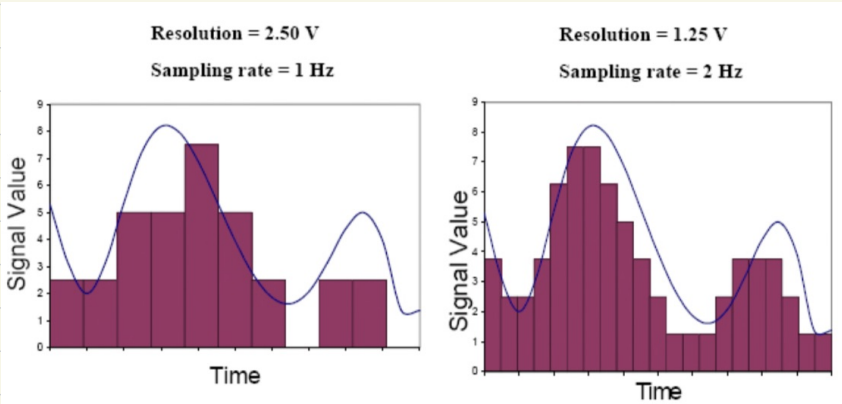
Atmega16 belongs to Mega category as

of pins = 40

Flash memory size = 16 KB

Why do we need Nyquist Law? What are the important points to achieve better accuracy in ADC? Is it possible to get the same analog signal after implementing ADC and DAC one after another? Justify your answer.

We need Nyquist law to get rid of Aliasing problem and increase the resolution and accuracy obtained from AD signals.



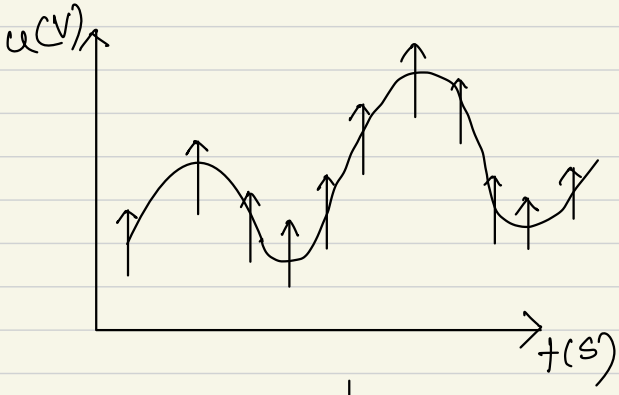
There are 2 ways to acheive better accuracy of A/D conversion

1. Increasing the resolution

   The more # of bits can be used to represent the signal. the more resolution it will be.
   But this is limited by SNR

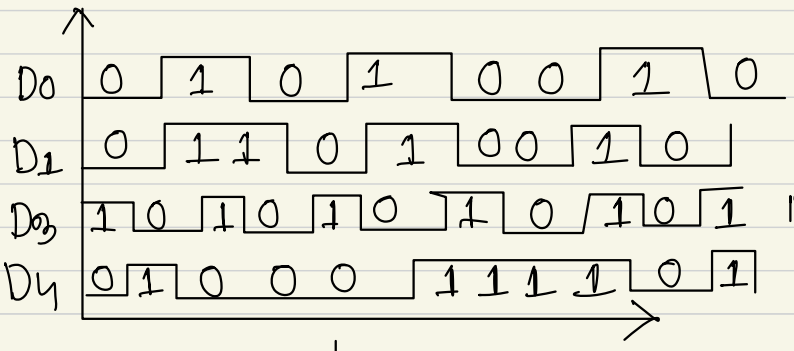
2. Increasing the sampling rate increases the max frequency that can be measured.

No, it is not possible to get the same signal.



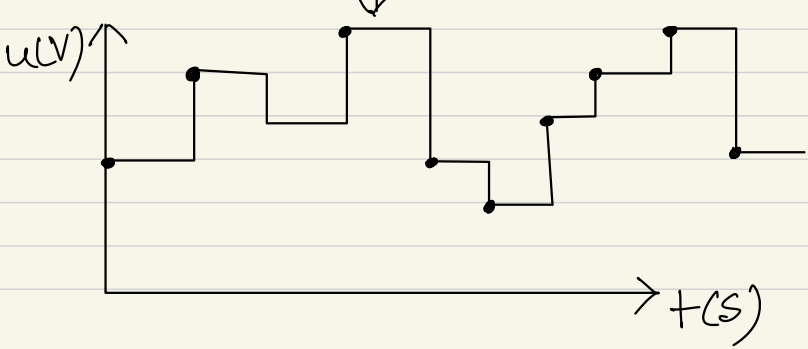
This can take
i) Infinity values
ii) change any time

ADC



i) Take 2 values
ii) changes at distinct times

DAC



i) Can only take predefined values

With respect to circuit design, Flash ADC and Weighted Sum DAC have an opposite similarity. Explain with appropriate diagrams.

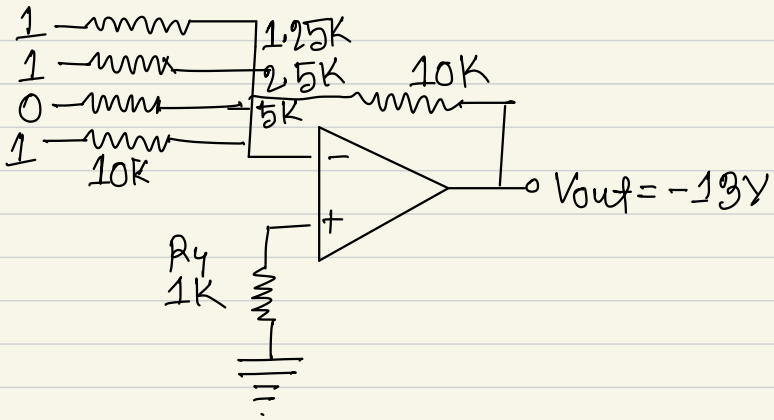
Flash ADC uses a comparator and a digital decoder to perform conversion. The analog voltage is compared to a set of. Vref and this operation is very fast

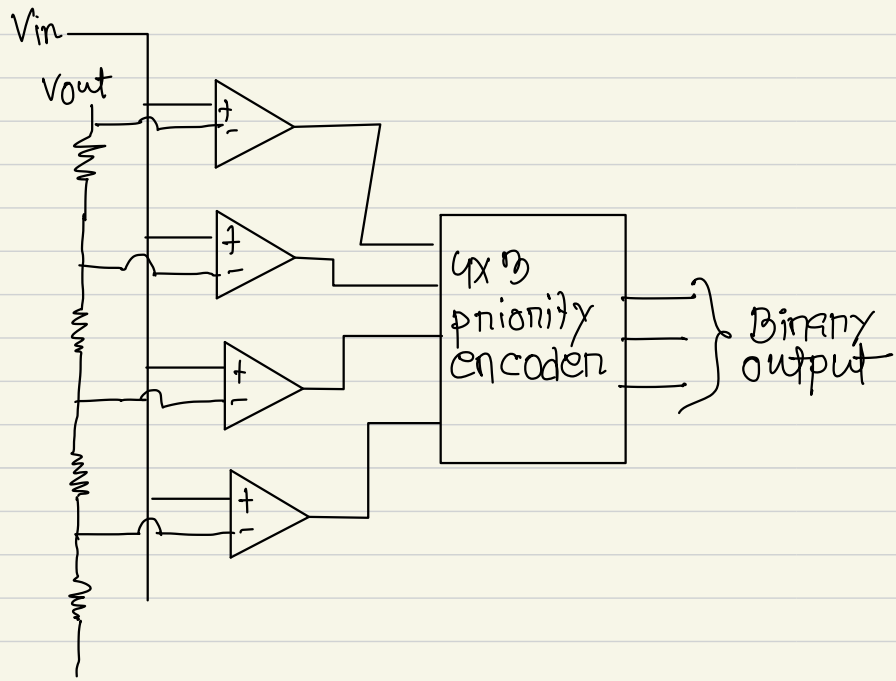
Weighted sum DAC uses a set of registers for each digit of the signal. The accuracy is defined by resistors and # of bits in the code

In flash ADC as the # of bits to represent the state of the signal increase the # of comparators exponentially increase

whereas in weighted sum ADC as digital signal goes from high to low the resistors multiplicative increase

So, in that sense both are similar


1
1
0
1
10K
1.25K
2.5K
5K
10K
Vout = -13V
Ry
1K


Vin
Vout
4x2
priority
encoder
Binary
output

Suppose, an 8086 microprocessor is asked to address a control word at the control register of 8255 PPI. Consider that in 8255, Port-A is in Mode-2, Port-B is in Mode-1 as an output port and Port-C is working for handshaking signals. Now, derive the binary values of A0 and A1 pins and draw the control word format for 8255 PPI.

It is given that,

MP is asked to address a control word at the control register

So,

A_1	A_0
1	1

Port A → Mode 2 (Bi-directional)

Port B → Mode 1 (Output port)

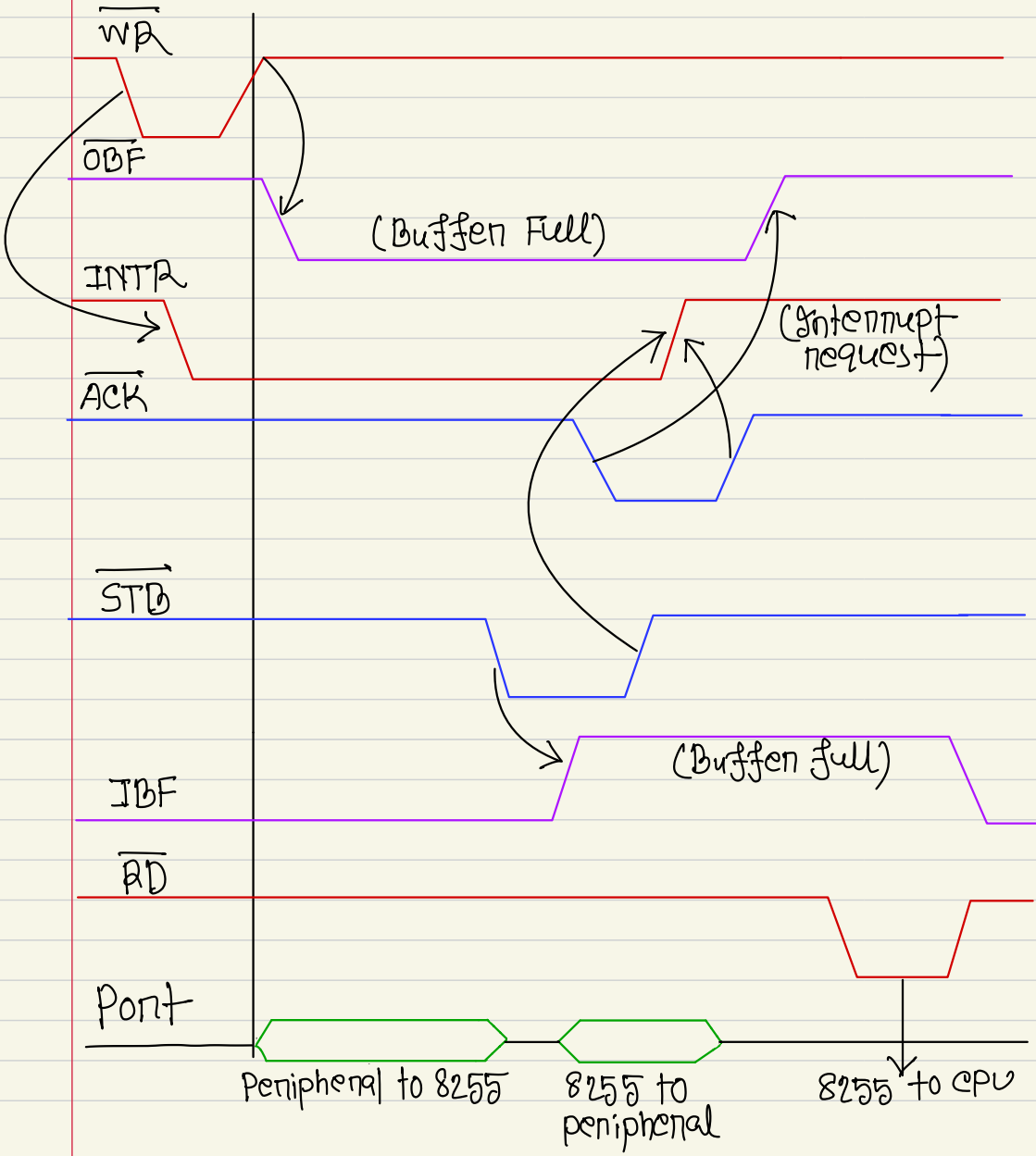
Port C → handshaking

Chip mode Mode PA PC4-PC7 Mode PB PC0-PC3

7	6	5	4	3	2	1	0
1	1	X	X	X	1	0	0

Group A Group B

Draw the sequential timing diagram for Port-A considering the handshaking and data signals (**Hint:** You have to consider the scenario of Question **3a**).



How does an asynchronous data transfer mode handle errors? Explain with a figure

Asynchronous data transfer adds

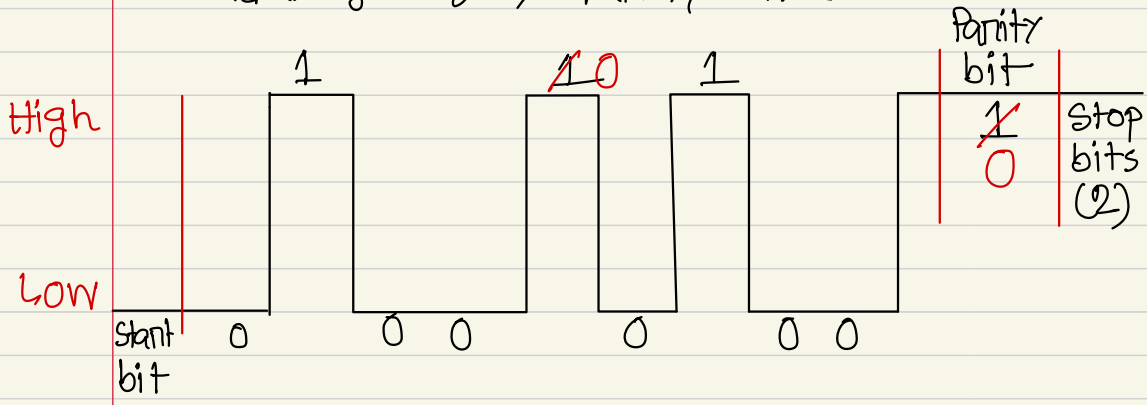
Start bit, Stop bits (2) and Parity Bit

Parity bit is used to handle errors in the transmission

Maintains odd parity

even # of 1's → Parity bit 0

odd # of 1's → Parity bit 1



if even # of bits are changed then error can't be detected here.

But odd # of changes can be handled using the parity bit

Mid 21

What is Aliasing Problem? How to solve it? Briefly explain the conditions to ensure accurate and precise A/D data conversion.

Aliasing problem

When input signal changes much faster than the sampling rate, this problem occurs.

Nyquist proposed the solution for this problem which is to use

Sampling freq = 2* max frequency

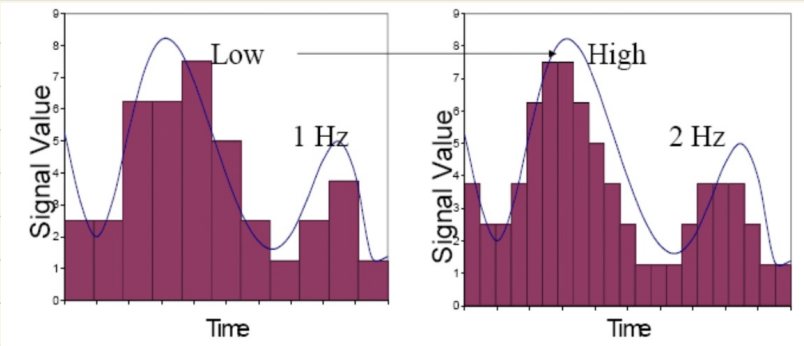
Two ways to improve the accuracy

1) Increasing the resolution

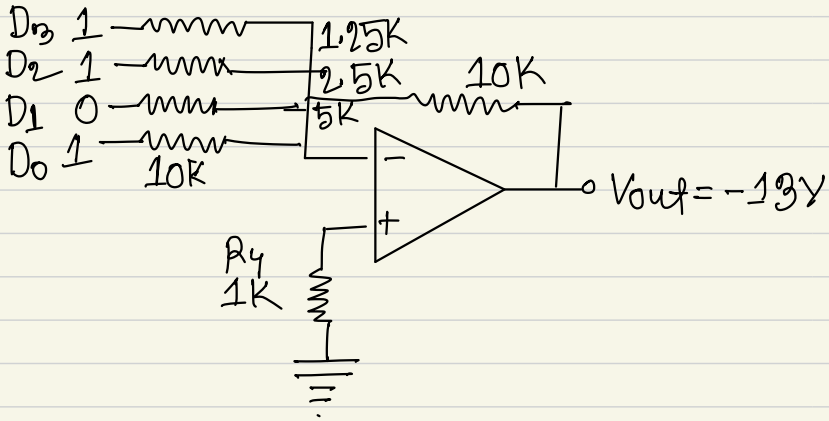
- Resolution is defined by # of discrete values the converter can produce
- This is limited by Signal-to-Noise Ratio
- Q = Vrange/2^n, Vrange is the range of analog voltages that can be represented.

11) Increasing the sampling rate

→ Increases the max frequency that can be measured



Differentiate among Weighted Sum and R-2R D/A converter.



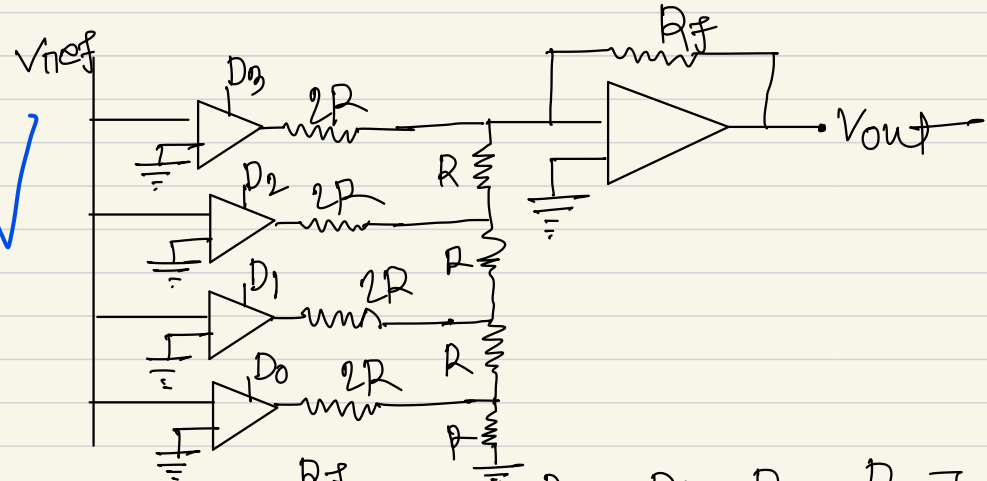
In weighted sum we have different weights

$$V_{out} = -(1\times 2^3 + 1\times 2^2 + 0\times 2^1 + 1\times 2^0)$$

MSB - LSB

The negative sign depends on the comparator

And for each bit we double the resistance making it non-scalable



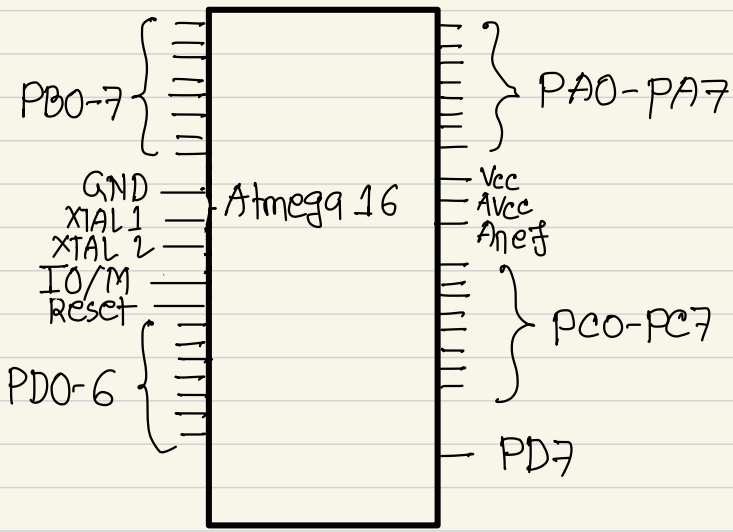
$$V_{out} = \frac{R_f}{R} V_{ref} \left[\frac{D_0}{16} + \frac{D_1}{8} + \frac{D_2}{4} + \frac{D_3}{2} \right]$$

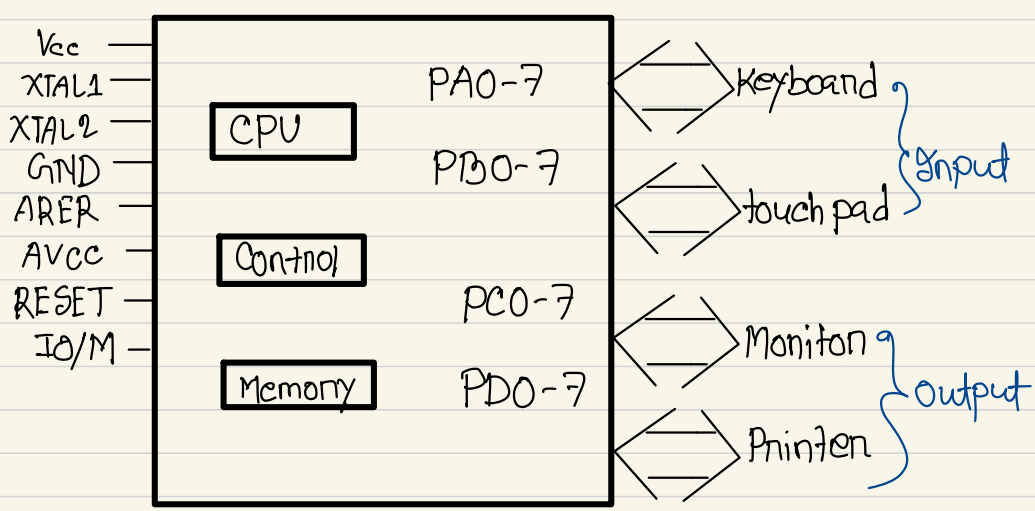
This only uses R resistor along the vertical axis as 2R along the horizontal axis.

If one more bit is added, it will easy to extend the circuit as we don't need to double the resistance

So, this gives us total modularity

Draw a block diagram using the specific ports pin numbers of ATMega16 microcontroller; where, you need to connect 2 input peripherals and 2 output peripherals.





Suppose, a control register of 8155 PPI has an address of `1Fh`. If following instructions are executed in an 8085 microprocessor system, then derive the all the port functionalities (i.e., including pins) of the 8155 PPI.

```
MVI A, ABh
OUT 1Fh
```

The control register has address 1Fh. We are passing control word to the control register

$$(AB)_{16} = (1010\ \ 1011)_2$$

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
1	0	1	0	1	0	1	1
Timer		IEB	IEA	PC3-PC5	PC0-PC2	PB	PA

Port A → Output

Port B → Output

PC0-PC2 → handshaking for port A

PC3-PC5 → Output

PC5	PC4	PC3	PC2	PC1	PC0
OUT	OUT	OUT	ACK_A	OBF_A	$INTR_A$

Interrupt disable → PA

enable → PB

Suppose, in a serial system total 30 frames (each having a size of 5 bytes) need to be transmitted. In case of asynchronous transmission 1 byte overhead occurs either for start or stop byte and 1-bit overhead occurs for error checking using parity bit. In contrast, for synchronous transmission 1 byte of synchronization overhead occurs after each 5 frame transmissions. Now, mathematically show the performance efficiency comparison between Synchronous Transmission and Asynchronous Transmission.

Asynch

Each frame = 5 byte + 2 byte + 1 bit
(Start/Stop) (Parity bit)

= (40+16+1) bit = 57 bit

Total data = (30 x 57) = 1710 bit

Synch

For 30 frames we need (30/5) = 6 synchronization frames

Total data = (30 x 5 x 8) + (6 x 8)

= 1248 bit

$$\text{Improvement} = \frac{1710 - 1248}{1710} \times 100\%$$

≈ 27% over asynchronous

Suppose, an 8086 microprocessor is asked to address the 48th 8255 and to write a control word at the control register of that 8255. Consider, Port-A is in Mode-1 as an output port, Port-B is in Mode-1 as an input port and Port-C is working for handshaking signals. Now, derive the binary values of A7 – A0 pins and draw the control word format for 8255.

locating control register

A_1	A_0
1	1

The 48th 8255 with indexing 0 will be,

$(47)_{10} = (101111)_2$

A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0
1	0	1	1	1	1	1	1

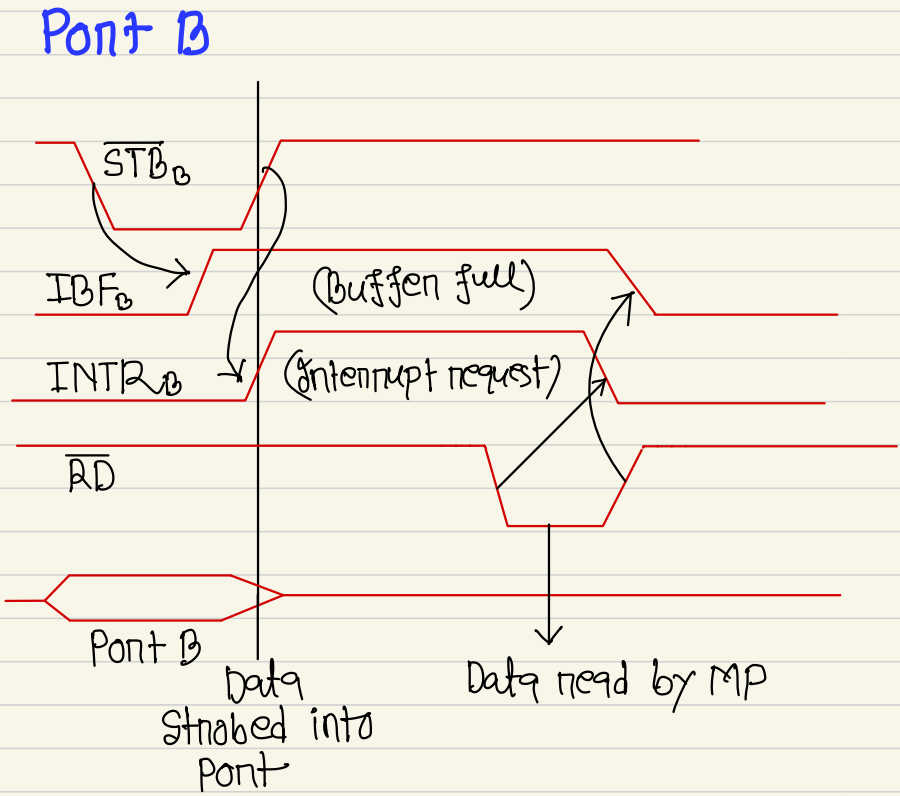
Port A → output → mode 1

Port B → input → mode 1

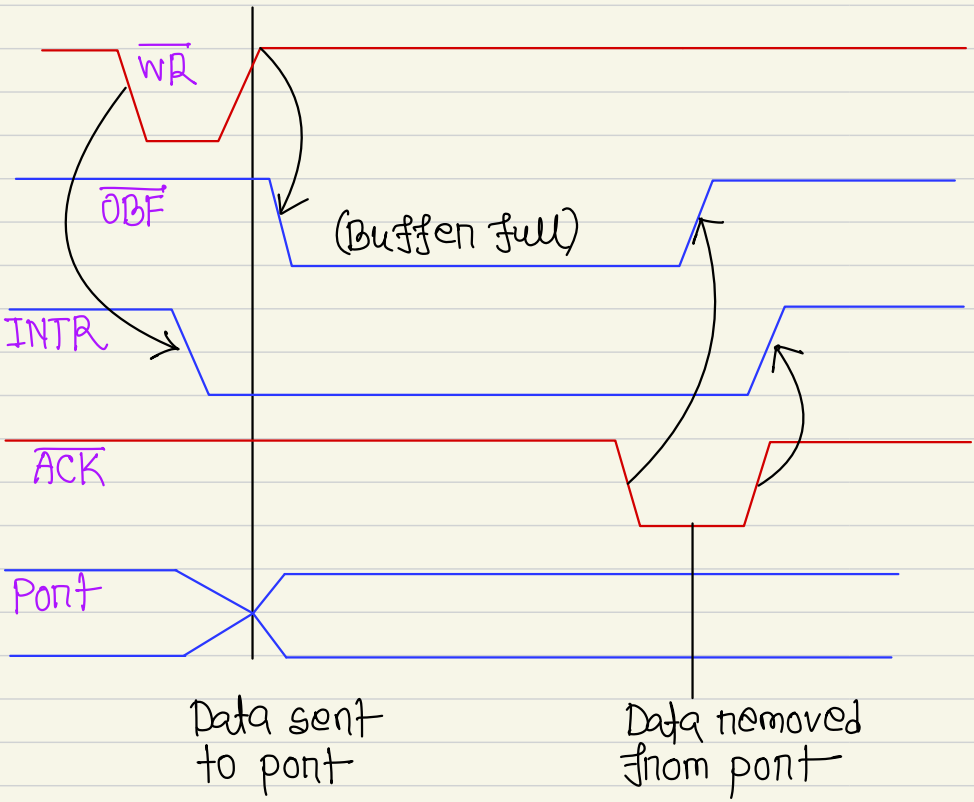
Port C → handshaking

Timer	Mode A		PA	PC7-PC4	Mode B	PB	PC3-PC0
7	6	5	4	3	2	1	0
1	0	1	0	0	1	1	1

Draw the sequential timing diagram for Port-A and Port-B considering the handshaking and data signals (consider the scenario of Question **3(a)**).



Port A


WR
OBF
(Buffer full)
INTR
ACK
Port
Data sent to port
Data removed from port

Differentiate among Delta-Sigma and Flash ADC.

Flash	Delta Sigma
Very fast	Very Slow
Lower resolution	higher resolution
Expensive	Low cost
less accurate	Highest accuracy among ADC's
Simple circuitry	Complex circuitry
Simple in operational theory	Complex
Output digital signal at once	Produces serial bit stream

Final 21

Write the pros and cons of the *Serial* and *Parallel* interface transmissions. How does the I^2C bus handle multi-master scenario?

Serial

- ✓ Cheaper
- ✗ Slower

Parallel

- ✓ Faster
- ✓ Simultaenous transmission
- ✗ Limited to small distances
- ✗ Requires seperate data lines
- ✗ Bits must be synchronized
- ✗ Expensive

a) What is meant by priority resolving for handling multiple interrupts? Which PIC is best suited with 8086 microprocessor? And why?

i) If priority is used to resolve an interrupt it is called priority resolving.

A priority resolving mechanism is used to determine which interrupt to handle 1st.

ii) 8259 PIC is used with 8086. It is a programmable device that can handle upto 8 interrupt signals

and provides priority resolving and masking capabilities.

It is widely known for reliability and compatibility with 8086.

c) Draw the control word format for the 8255A PPI when Port-A is connected with a 7-Segment display and other ports are also in output mode.

Final 18-19

b) Differentiate between *Asynchronous*, *Synchronous* and *Isochronous* data transfer

	Synchronous	Asynchronous
Process	Transmit data in form of chunks	Transmits 1 byte at a time
Speed	Fast transmission	Slow transmission
Cost	Expensive	Cheap
Interval	Time interval is constant	Time interval is irregular
Sender and receiver	Must be synchronized with same clock	No need for synchronization
Overhead	Synchronization bit	Start, Stop and parity bits

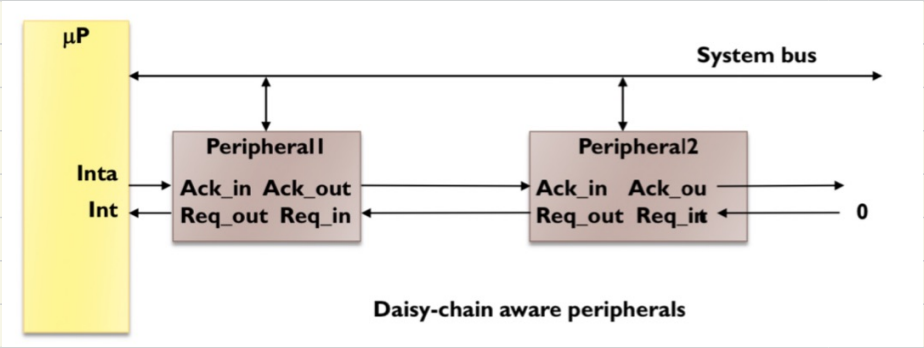
a) Write the basic differential features between 8155 and 8255 Programmable Peripheral Interface (PPI).

	8155	8255
Pin Config	24 pin DIP	40 pin DIP
# of I/O ports	2 programmable 8 bit, 1 programmable 6 bit	3 programmable 8 bit
Modes of operation	2 modes mode 0, mode 1	3 modes mode 0, mode 1, mode 2
Timer	Built-in 14 bit timer	External timer
Chip select	1 chip select signal	2 chip select signals
Interrupt Capability	Enabled	Disabled
RAM	Available	Not available
Port A	Uni-directional I/O port	Bi-directional I/O port
MP	Compatible to work with 8085	Compatible to work with 8086

b) Write short notes on:
   i. Daisy-Chain Architecture
   ii. Multi-Level Bus Architecture

i) Daisy-Chain Architecture

- Arbitration done by peripherals
- One peripheral connect to resource, all others connected upstream
- Req. flows downstream to resource and ACK flows upstream.
- Priority based on distance to resource
- Doesn't support rotating priority
- Single point of failure



11) Multi-level Bus Architecture

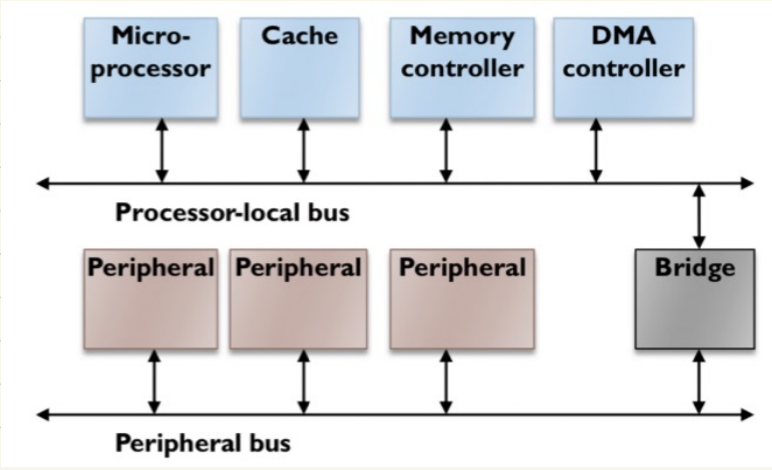
- Processor-local bus connects MP, cache, MC and DMA controller.

  High speed, wide, frequent communication

- Peripheral bus connects all peripherals.

  low speed, narrow, less frequent comm^{n}.

- Bridge connects processor-local bus and peripheral bus

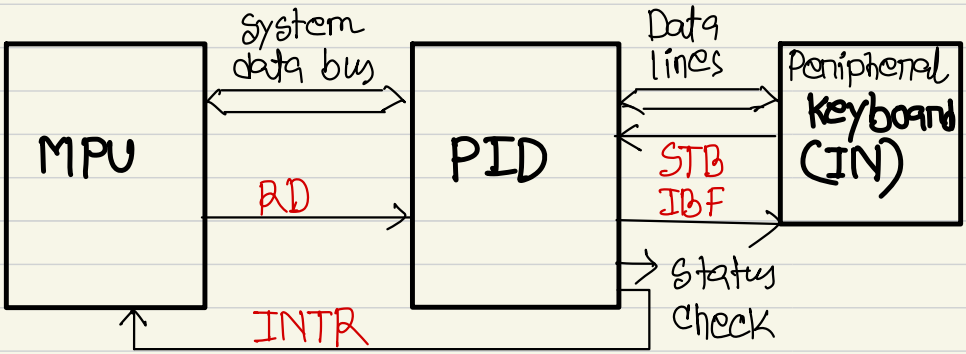


Explain about the handshake signals of 8155 Programmable Peripheral Interface (PPI).

There are 2 types of handshake signals

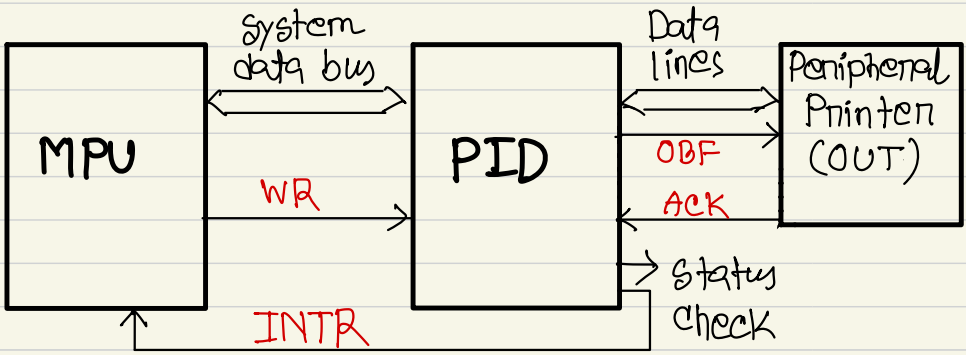
1) Input Handshake Signals

- Peripheral puts data in data-line and sends STB
- Device informs peripheral that IN port is full and sends IBF
- Either MPU checks status or Device interrupt to MPU for reading data from device

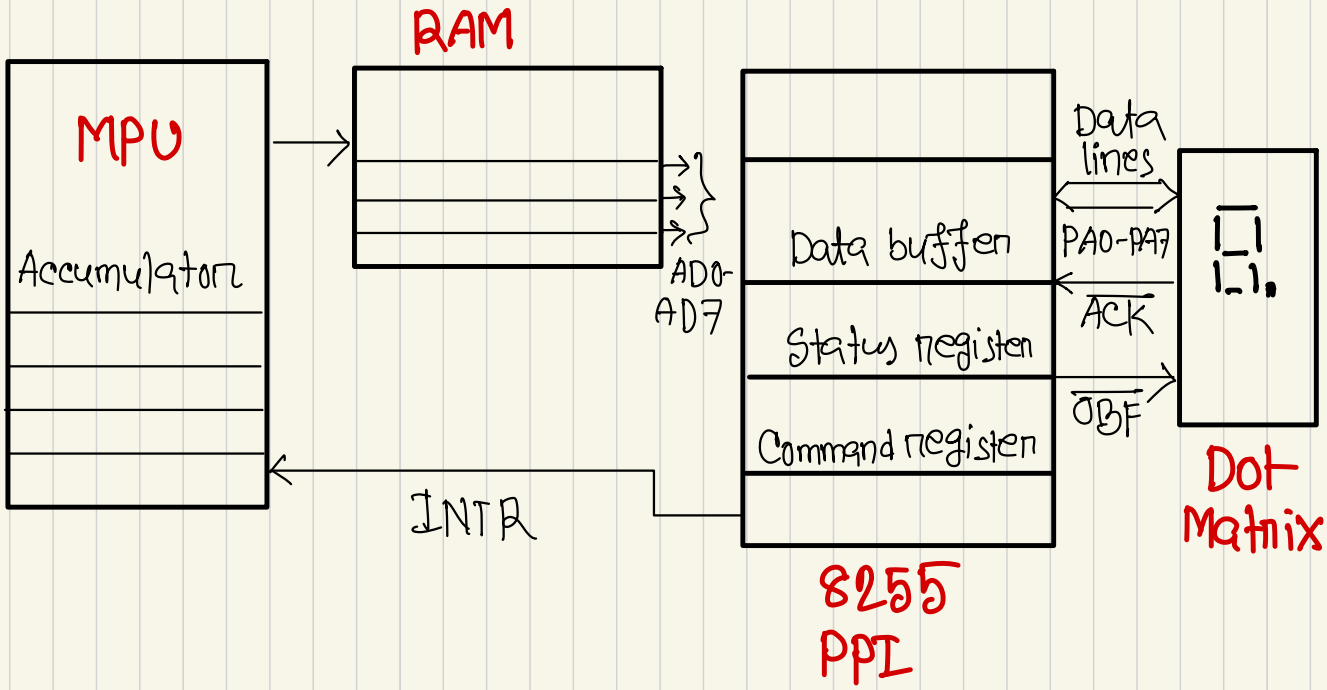


ii) Output handshake signals

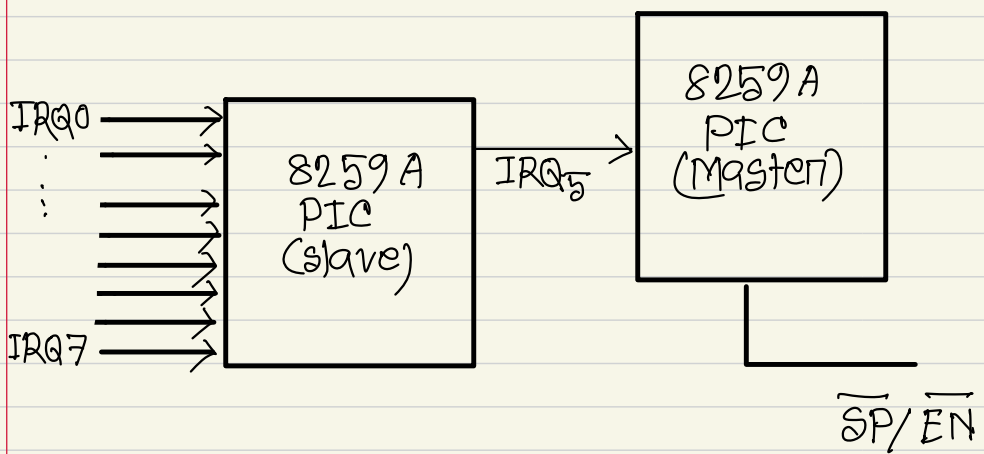
- → MPU writes byte by sending WR
- → Device informs peripheral by sending OBF
- → Peripheral sends ACK to the device
- → Device interrupt MPU by sending INTR on MPU checks status



In order to connect a Dot-Matrix Display with an 8086 Microprocessor system, how can you interface using a single 8255 PPI? Just draw the interfacing diagram.



How does 8259 PIC can handle 64 Interrupt levels? Explain with necessary diagram.



IRQ0 → highest **IRQ7** → lowest

CAS2-CAS0 is used to identify a particular slave 8259A to be accessed.

CAS2	CAS1	CAS0
0	0	0
0	0	1
0	1	0
⋮	⋮	⋮
1	1	1

$\overline{SP}/\overline{EN}$ is used to enable the master/slave configuration in an 8259A PIC

Briefly explain about the operation of Register sets of 8259 PIC

There are 4 registers in 8259 PIC

1) IMR (Interrupt Mask Register)

Used to mask on enable interrupts from specific devices connected to PIC

11) IRR (Interrupt Request Register)

Indicates which interrupt requests are currently pending

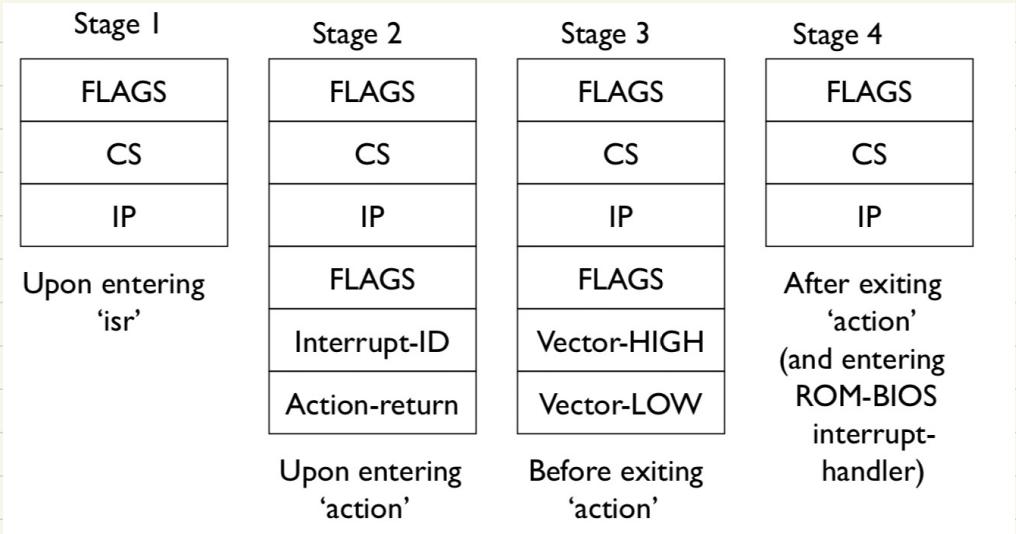
111) ISR (Interrupt Service Register)

Indicates which interrupts are currently being serviced

1v) ICR (Interrupt Command Register)

Used to set mode of operations and priority of interrupts.

Draw the STACK operation states during an Interrupt.



MID 18-19

Differentiate between Microcontroller and Microprocessor.

	Microcontroller	Microprocessor
Components	CPU, RAM, ROM, I/O and timer are all on single chip	CPU is stand-alone
Purpose	Single purpose	General purpose
Cost	Cost effective	expensive
Processing power	Lower	Higher
Size	Smaller	Larger
Memory	Internal- limited size	External- Flexible
I/O	Programmable digital and analog I/O pins	External peripherals with I/O pins

What are the steps involved in Analog-to-Digital (A/D) data conversion? Briefly explain the conditions to ensure accurate and precise A/D data conversion.

1)

2 step process

i) Quantizing :

→ breaking down analog value in a set of finite states

→ Quantization size,

$$Q = \frac{V_{max} - V_{min}}{N}$$

where, N = # of possible states

$$N = 2^n$$

where, n = # of bits in the A/D conventer

ii) Encoding

→ Assiging a digital word/number to each state and matching it to the input signal

Write short notes on:

i. Microprocessor controlled data transfer
ii. Peripheral controlled data transfer

1) Microprocessor Controlled

→ Microprocessor directly controls the flow of data between the MP and peripheral

→ Sends signal to peripheral to transfer data or receive data from MP

→ Different techniques such as

a) Unconditional
b) Polling
c) Interrupt
d) READY signal
e) Handshake signal

ii) Peripheral Controlled

- Peripheral controls the flow of data
- Peripheral initiates data transfer by sending signal
- Used when peripheral is faster than MP
- **Example**

  DMA
- Data transferred at high speed without MP

Suppose, an 8086 microprocessor is asked to address the 15th 8255 and to write a control word at the control register of that 8255. Consider, Port-A is in Mode-2, Port-B is in Mode-1 as an output port and Port-C is working for handshaking signals. Now, derive the binary values of A7 – A0 pins and draw the control word format for 8255.

locating control register

A_1	A_0
1	1

The 15th 8255 with indexing 0 will be,

$(15)_{10} = (001111)_2$

A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0
0	0	1	1	1	1	1	1

Port A → Bi-directional → mode 2

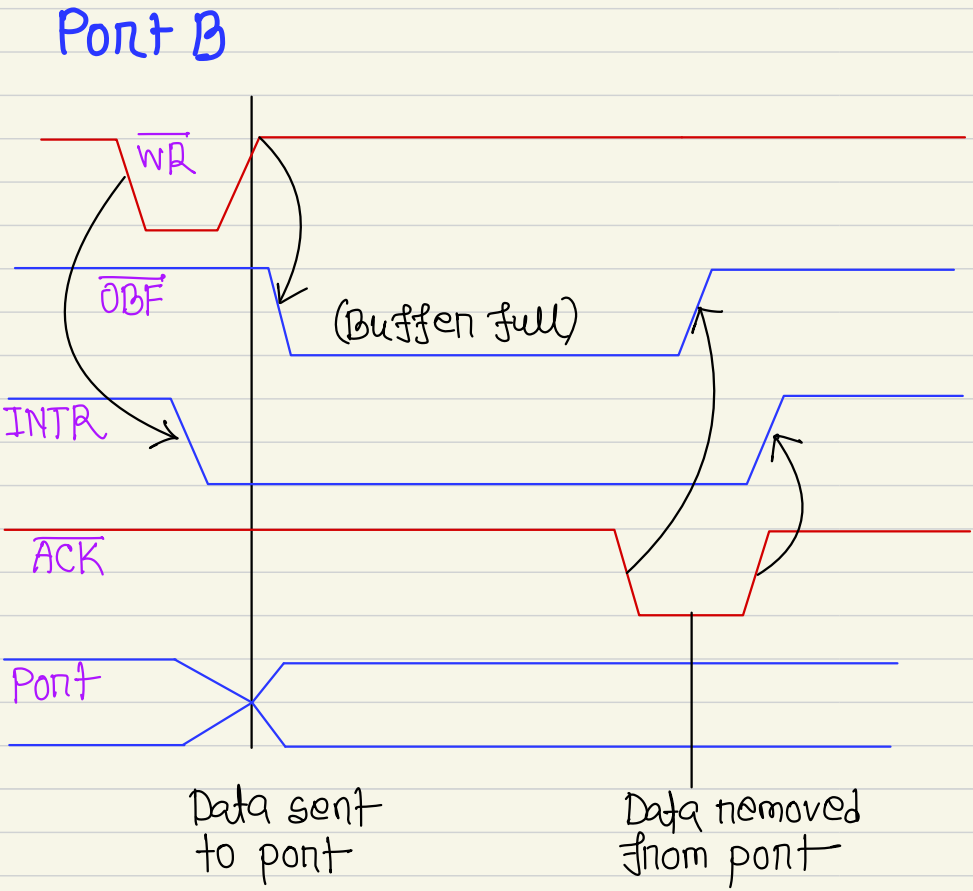
Port B → Output → mode 1

Port C → handshaking signals

Timer	Mode A		PA	PC7-PC4	Mode B	PB	PC3-PC0
7	6	5	4	3	2	1	0
1	1	X	X	X	1	0	0

Group A Group B

Draw the sequential timing diagram for Port-B considering the handshaking and data signals (consider the scenario of Question 3.b).



Write down the features of 8255 PPI.

i) Compatible to work with 8086 MP

ii) General purpose I/O interfacing

iii) 24 pins for I/O that are programmable in groups of 12 pins

iv) 3-8 bit I/O ports

v) 40 pin dual-inline-package

vi) Has 3 operation modes

vii) In mode 2, Port A can be used for Bidirectional I/O

Final 17-18

'Microprocessor is generalized and microcontroller is specific' – Explain how.

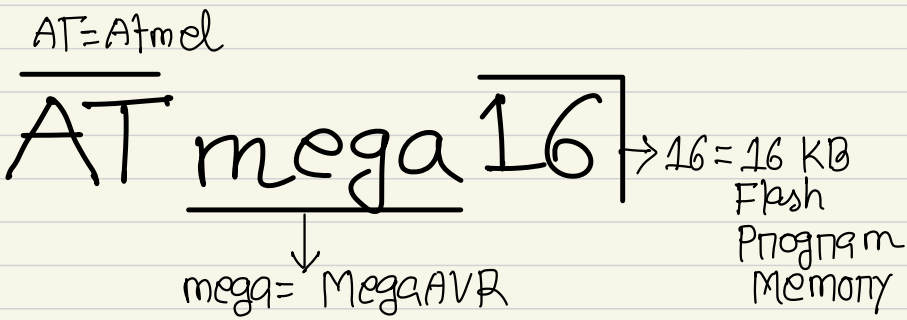
Microprocessors are often considered to be more generalized than microcontrollers because they are designed to handle a wide range of tasks and are commonly used in general-purpose computing applications.

Microprocessors typically require external components, such as memory, input/output (I/O) ports, and peripherals, to perform their functions. They are designed to execute instructions and process data that is stored in memory, and they can be programmed to perform a wide range of tasks, from basic arithmetic operations to complex algorithms.

On the other hand, microcontrollers are considered to be more specific than microprocessors because they are designed for a specific application, such as controlling motors, managing sensors, or processing audio signals.

Microcontrollers are self-contained, meaning that they include integrated memory, I/O ports, and peripherals that are specific to the task they were designed to perform. They are commonly used in embedded systems, which are specialized computer systems that are integrated into other devices or products.

What is the naming convention of ATMEL microcontroller? Define and distinguish between Tiny, Mega and XMega AVRs (Advanced Virtual RISC).



Tiny AVR

Less memory, small size, suitable for simple applications

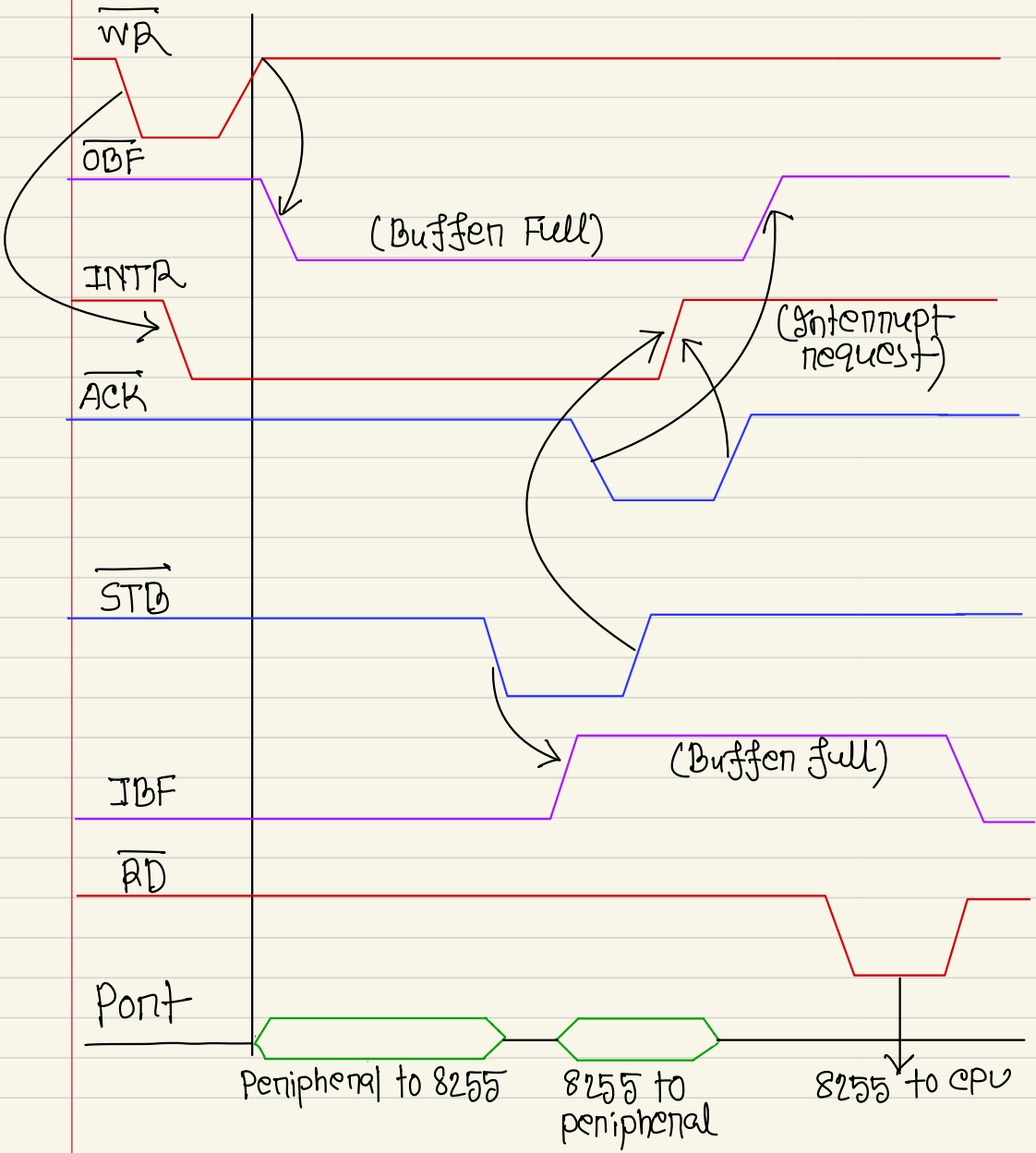
Mega AVR

Good amount of memory, higher # of inbuilt peripherals, suitable for moderate to complex applications

XmegaAVR

Large memory, high speed, complex applications

Draw and explain the timing diagram of 8255 PPI mode 2 signaling.



Write the taxonomy of models of transfer in Peripherals and Interfacing along with their features.

Mid 17-18

'*Data transfer with Interrupt is a microprocessor controlled approach, whereas data transfer with Direct Memory Access (DMA) is a peripheral controlled approach*' – True or False? Justify your answer.

True